APPENDIX 2

Domain-Specific Languages

Domain-specific languages are a popular topic these days. They overlap somewhat with metaprogramming, so you'll probably want to know a thing or two about them.

The Case for Domain-Specific Languages

Are you old enough to remember *Zork*? It was one of the first "text adventures": text-based computer games that were popular in the early 1980s. Here are the first few lines from a game of *Zork*:

```
West of house
You are standing in an open field west of a
white house, with a boarded front door.
You see a small mailbox here.
⇒ open mailbox
Opening the small mailbox reveals a leaflet.
⇒ take leaflet
Taken.
```

Suppose you have to write a text adventure as your next job. What language would you write it in?

You'd probably pick a language that's good at manipulating strings and supports object-oriented programming. But whatever language you chose, you'd still have a gap between that language and the problem you're trying to solve. This probably happens in your daily programming job as well. For example, many large Java applications deal with money, but `Money` is not a standard Java type. That means each application has to reinvent money, usually as a class.

In the case of our adventure game, you have to deal with entities such as rooms and items. No general-purpose language supports these entities